

NetComms 2 Modules 1 and 2 Reviewer

CS0016 Network and Communications 2 | DevNet Associate. Blue boxes are labeled memory aids for study, not slide text.

Module 1: Course Introduction

Module objective: Use basic Python programming and Linux skills. The topics prepare you to install a virtual lab environment, manage the Linux file system and permissions, and use basic Python.

Course Overview

The DevNet Associate course covers software development, networking fundamentals, and automation through five course modules.

Course Module	What It Covers
Understanding and Using APIs	APIs, their benefits, and troubleshooting
Software Development and Design	Main software development concepts and tools for quality code
Network Fundamentals	Networks, devices, protocols, and connectivity troubleshooting
Infrastructure and Automation	Managing infrastructure with automation instead of manual setup
Cisco Platforms and Development	Data centers and networking, including data models and security

Your Lab Environment

Virtualization lets virtual computers run within a physical computer. These virtual computers are **virtual machines (VMs)**. A VM is a **guest**; the physical computer is the **host**. A modern computer and operating system can run VMs.

Watch Out

Guest = virtual machine. Host = physical computer. Do not swap the machine running inside with the machine it runs on.

The Three Labs

Lab	Parts
Install the Virtual Lab Environment	1. Prepare a computer for virtualization; 2. Explore the DEVASC VM GUI; 3. Create lab environment accounts; 4. Install Webex Teams on your device.
Linux Review	1. Launch the DEVASC VM; 2. Review command syntax navigation; 3. Review file management; 4. Review regular expressions; 5. Review system administration.
Python Programming Review	1. Launch the DEVASC VM; 2. Start Python and VS Code; 3. Review data types and variables; 4. Review lists and dictionaries; 5. Review the input function; 6. Review if, for, and while functions; 7. Review methods for file access.

Linux for DevNet

Linux is widely used in servers, IoT devices, networking equipment, smartphones, and other devices. **All coding labs in this course use a Linux-based VM.** If the Linux Review lab is difficult, **Linux Unhatched** is a free, online, self-paced remedial course.

Why Python?

1. **Easy to learn:** learning takes less time than with many other languages.
2. **Easy to use:** you can write new software faster.
3. **Easy to obtain, install, and deploy:** Python is free, open, and multiplatform.

Python also provides a foundation for learning other languages, including C++, Java, and C. If the Python Programming Review lab is difficult, **Python Essentials** on the Student Resources page is free, online, and self-paced.

Memory Aid

M1 readiness chain: install the VM, review Linux, then review Python. Linux Unhatched supports the Linux review; Python Essentials supports the Python review.

Module 2: The DevNet Developer Environment

Module objective: Implement a development environment using DevNet resources. Explain how DevNet encourages communities of network programmers and investigate its online resources.

What Is DevNet?

DevNet is a fully integrated developer program with a website, interactive developer community, coordinated developer tools, integrated discussion forums, and sandboxes.

Feature	Role
Learning Labs	Self-paced tutorials, from basic coding to REST APIs with different technologies
Sandboxes	Production-like environments for development and testing
Code Exchange	Sample code written by other developers
Developer Support	Developer issue support through tickets, live chats, and forums
Developer Documentation	Central location for product developer API documentation

Getting Started with DevNet Resources

DevNet is a starting point for Cisco APIs, with API documentation, education, and developer support. The home page at **developer.cisco.com** shows these tiles: Start Now, Learning Tracks, Video Course, Sandbox, Code Exchange, and Ecosystem Exchange.

The pictured platform icons are **IoT, Cloud, Networking, Data Center, Security, Mobility, Open Source, Collaboration, and Services**.

DevNet Learning Labs

Learning Labs provide tutorials on engineering technologies, programming languages, model-driven programmability, REST APIs, Python, and JavaScript. They also provide a walk-through for using a DevNet Sandbox already configured with Cisco platforms.

They enable practice by setting up a development environment on a local computer and offer tutorials across coding, collaboration, IoT, data center, mobility, and networking topics.

DevNet Sandbox

Sandboxes are environments for hands-on exploration of software and APIs. The slide picture shows these sandbox cards:

Sandbox Card	Detail Shown
Firepower Management Center	REST API; reservation only
Meraki Small Business / Meraki Enterprise	Meraki sandbox cards
Network Assurance Engine	Always-on

Watch Out

Learning Labs = guided tutorials and walk-throughs. Sandbox = the environment where you explore software and APIs. A Learning Lab may guide you through a Sandbox.

The Three DevNet Exchanges

Exchange	What It Holds
Automation Exchange	Network automation use cases with solutions and toolkits: listing data, adding configurations, and activating policies across domains, users, or devices. Listings cover tool sets such as Ansible or Puppet and campus or branch, data center, or service provider infrastructure.
Code Exchange	Source code and tools. It uses the GitHub API and human moderators to categorize and display hundreds of related repositories.
Ecosystem Exchange	More than 1,500 solutions across technologies, industries, and geographies to start solution design and development.

Watch Out

Automation = use cases and toolkits; Code = repositories; Ecosystem = solutions. The GitHub API and moderators belong to Code Exchange, not Automation Exchange.

Memory Aid

A-C-E: Automation gives a use case, Code gives source code, Ecosystem gives a solution catalog.

DevNet Developer Support

Support helps troubleshoot integrations, API connections, and developer use cases on Cisco products. The three ways to reach it are **log a ticket, post to a community forum, and access a Webex Teams space**. The slide directs readers to **developer.cisco.com/support**.

The **Knowledge Base** contains troubleshooting articles. A **Support Case** provides one-on-one support with a one-business-day response. The **Cisco Developer Community forums** are available through **devnetsupport.cisco.com**, then Community.

Support Option	Cost	What The Slide Shows
Knowledge Base	Free	Troubleshooting articles
Community Forum	Free	Community discussion
Chat with DevNet	Free	Chat support
Case-Based Ticket	USD 250 for 1 ticket; USD 950 for 5 tickets	One-on-one support; one-business-day response

Module 2 Summary

DevNet starts developers with Cisco APIs. Its online resources include Learning Labs, video courses, Sandbox, Exchanges, and developer support. The **Explore DevNet Resources** lab has two parts: **find and navigate a Learning Lab**, then **explore more resources**.

One-Page Cram Sheet

Recall Prompt	Answer
M1 objective	Basic Python and Linux; install VM, manage Linux files and permissions
Course modules	APIs; Software Development and Design; Network Fundamentals; Infrastructure and Automation; Cisco Platforms and Development
Guest / host	Guest = VM; host = physical computer
Three M1 labs	Install Virtual Lab Environment (4 parts); Linux Review (5); Python Programming Review (7)
Linux	All coding labs use a Linux-based VM; Linux Unhatched if review is difficult
Python	Easy to learn; fast to write; free, open, multiplatform; foundation for other languages; Python Essentials if review is difficult
M2 objective	Implement a development environment with DevNet resources
Five DevNet features	Learning Labs; Sandboxes; Code Exchange; Developer Support; Developer Documentation
Labs / Sandbox	Tutorials and walk-throughs / hands-on testing environment
Three Exchanges	Automation = use cases; Code = GitHub-linked repositories; Ecosystem = 1,500+ solutions
Automation examples	List data, add configurations, activate policies; Ansible or Puppet; campus/branch, data center, service provider
Support access	Ticket, community forum, Webex Teams space
Support cost	Knowledge Base, forum, chat: free. Case ticket: USD 250 for 1 or USD 950 for 5; one business day response

Memory Aid

Sort the easy swaps: guest = VM; Labs = guided learning; Sandbox = practice environment; Automation = use cases; Code = repositories; Ecosystem = solutions.